Student Resource Guide

Generative AI for Beginners

Section 6: Training and Fine-Tuning | Lecture 6.3: Data Requirements and Quality

Learning Objectives

- Identify the data quantity requirements for pre-training and fine-tuning across different model types
- Describe the key characteristics of high-quality training data: accuracy, diversity, relevance, and cleanliness
- Explain the major data quality problems (bias, copyright, PII, misinformation) and their impact on model behaviour
- Apply data curation best practices to fine-tuning dataset preparation
- Evaluate the ethical dimensions of training data provenance and consent

Introduction

The AI field has a guiding principle that deserves to be treated as foundational: 'Garbage in, garbage out.' The quality and composition of training data are the single greatest determinant of what a model learns, what it gets right, what it gets wrong, and what harmful behaviours it may exhibit. No amount of architectural sophistication, compute investment, or prompt engineering can fully compensate for fundamental problems in the training data. Conversely, high-quality, well-curated data enables even relatively modest models to achieve exceptional performance.

Data quality is not a single dimension. Training data must simultaneously be accurate (containing correct information), diverse (covering the full target distribution), relevant (aligned with the intended application), consistent (formatted and labelled uniformly), and clean (free from toxic content, PII, duplicates, and copyright-infringing material). These dimensions sometimes create tension: maximising diversity may introduce noise; maximising quality through curation may reduce diversity. Understanding how to navigate these trade-offs is a core practical skill in AI development.

For practitioners who are building fine-tuning datasets rather than pre-training from scratch, these principles apply at a smaller but no less important scale. The 5,000 examples you curate for fine-tuning will have an outsized influence on model behaviour—far more than 5,000 examples in a trillion-token pre-training corpus. Data curation for fine-tuning requires more careful judgement per example, not less.

Key Concepts

Data Quantity Requirements by Training Phase

Definition	The minimum and practical training data volumes differ dramatically by training phase: pre-training requires hundreds of billions to trillions of tokens (language) or billions of images; fine-tuning typically requires thousands to hundreds of thousands of examples; RLHF requires tens to hundreds of thousands of human preference pairs.
Business Context	Understanding data quantity requirements informs project scoping and cost estimation. Fine-tuning feasibility assessment begins with a realistic inventory of available high-quality labelled examples. If fewer than 1,000 domain-specific examples are available, fine-tuning may not produce sufficient improvement over prompting to justify the development effort.
AI Practitioner relevance	When planning fine-tuning projects, practitioners should establish a data collection and annotation pipeline before committing to a fine-tuning approach. The data collection phase is often the longest and most expensive part of a fine-tuning project—far exceeding the training compute cost.
Real-World Example	A legal tech company estimated that fine-tuning their contract review model required 50,000 labelled clause extraction examples. Annotation at $0.50/example (legal domain expertise required) totalled $25,000 in annotation cost—compared to $500 in GPU compute cost for the fine-tuning run itself. Data annotation was 50x the training cost.

Training Data Bias and Its Consequences

Definition	Systematic imbalances in training data that cause models to perform differently across demographic groups, cultural contexts, or topic domains. Bias sources include: sampling bias (certain groups over/under-represented), measurement bias (labels

	that reflect human annotator biases), historical bias (data reflecting past discriminatory practices), and representation bias (certain perspectives consistently absent).
Business Context	Biased models create business, legal, and reputational risk. A recruitment AI that systematically downscores certain demographic groups may violate equal employment law. A customer service AI that provides worse responses to queries in certain dialects creates differential customer experience. Bias assessment is a legal and reputational necessity, not just an ethical consideration.
AI Practitioner relevance	Practitioners should conduct bias assessment on any model used in consequential decisions (hiring, lending, medical, legal, customer service). Assessment should cover: demographic subgroup performance parity, response quality consistency across languages and dialects, and intersectional analysis (where multiple attributes combine to create compound disadvantage).
Real-World Example	An AI hiring screening tool trained on historical hiring decisions inherited the hiring biases of those decisions: resumes from graduates of historically Black colleges and universities (HBCUs) scored lower because historically fewer HBCU graduates had been hired—not because of qualification differences. Amazon's internal AI recruiting tool was scrapped in 2018 after this type of bias was discovered.
Key Risks	Bias mitigation is difficult because bias is often embedded in subtle statistical patterns distributed across training data. Techniques (resampling, reweighting, adversarial debiasing) reduce but rarely eliminate bias. Continuous monitoring of production model performance across demographic subgroups is necessary even after debiasing efforts.

Data Deduplication and Quality Filtering

Definition	Data preprocessing steps that remove duplicate content and filter out low-quality

	examples before training. Deduplication removes exact and near-duplicate content that would cause models to memorise rather than generalise. Quality filtering removes spam, gibberish, toxic content, and low-information-density text using heuristics and classifier-based approaches.
Business Context	Deduplication and quality filtering directly improve training efficiency: eliminating near-duplicates means the model learns from more distinct examples per training step. It also reduces memorisation risk—models that see the same text many times are more likely to memorise and regurgitate it, raising copyright and PII concerns.
AI Practitioner relevance	For fine-tuning dataset preparation, practitioners should apply: exact deduplication (remove identical examples), near-deduplication (fuzzy matching for structurally similar examples with minor variations), length filtering (remove very short examples that lack context), and relevance filtering (assess whether each example is representative of production queries).
Real-World Example	Common Crawl (the primary pre-training data source for most LLMs) is estimated to contain 25–50% near-duplicate content before deduplication. GPT-3's data processing pipeline reduced the Common Crawl dataset from petabytes to ~570GB through deduplication and quality filtering—retaining approximately 10% of raw data volume.

PII and Copyright in Training Data

Definition	Personally Identifiable Information (PII) in training data (names, email addresses, phone numbers, national IDs) can be memorised and reproduced by models, creating privacy violations. Copyright-protected text in training data raises legal questions about whether model training constitutes copyright infringement and whether model outputs can infringe copyright through near-reproduction.
Business Context	Several generative AI companies face

	ongoing litigation over training data: The New York Times sued OpenAI in December 2023, alleging that ChatGPT reproduces copyrighted NYT articles verbatim. This legal uncertainty affects both training data procurement and the legal status of AI outputs. Practitioners should maintain records of training data sources and provenance.
AI Practitioner relevance	For fine-tuning data, practitioners should: strip PII from examples before use, verify consent or licence for any copyrighted content, avoid using data from sources with Terms of Service prohibiting AI training (many platforms now include this), and document data provenance and any licence agreements for audit purposes.
Real-World Example	A company fine-tuning a legal AI on case law text confirmed that the legal text database they intended to use had Terms of Service prohibiting AI training use. They negotiated a specific AI training licence addendum before proceeding. This documentation protected them when the database provider later attempted to audit AI training customers.

Real-World Applications

Data quality principles apply across all AI development contexts—from enterprise fine-tuning projects to evaluating third-party AI tools.

Fine-Tuning Dataset Curation Pipeline: A structured data curation pipeline for fine-tuning projects should include: data collection (sourcing from production logs, domain experts, or annotation), preprocessing (normalisation, deduplication, length filtering), quality review (human annotation of a sample to assess quality distribution), PII and copyright screening, format standardisation, and held-out test set creation before any model training begins.

Bias Assessment in Production AI Systems: Organisations deploying AI in consequential decision-making should conduct quarterly bias audits: sample model outputs across demographic subgroups relevant to the use case, measure performance disparity, investigate root causes in training data, and document findings. Bias assessments should be performed by teams with demographic expertise, not just data scientists.

Data Provenance Documentation for Regulatory Compliance: The EU AI Act requires high-risk AI systems to document training data sources, characteristics, and quality measures. Establishing data provenance documentation at the point of data collection is dramatically cheaper than reconstructing it for regulatory compliance after deployment. Build provenance logging into the data pipeline from the start.

Synthetic Data Generation for Training Data Augmentation: When real-world training data is scarce, expensive to annotate, or privacy-restricted, synthetic data generated by existing AI models can supplement fine-tuning datasets. This requires careful quality validation (the synthetic examples must accurately represent target behaviours) and bias assessment (synthetic data inherits the biases of the generating model).

Best Practices

Invest more time in data curation than in hyperparameter tuning. Data quality improvements compound: better training data improves every subsequent use of the model. A 2-week data curation effort typically outperforms a 2-week hyperparameter tuning effort in terms of final model quality. Front-load data quality work.

Create diverse examples that cover edge cases, not just common cases. Models trained primarily on common, easy examples fail on edge cases that are rare in training data. Deliberately sample and annotate examples from the tail of your distribution—unusual phrasings, rare topics, challenging formatting—to improve robustness.

Establish a 'golden dataset' for consistent evaluation. Curate a fixed, human-verified evaluation set before any training begins. This set should not change between experiments—it is your measurement instrument. Changing the evaluation set invalidates comparisons between model versions.

Apply the same data quality standards to synthetic data as to real data. Synthetic data used for fine-tuning can introduce systematic errors if the generating model has gaps or biases. Review synthetic examples with domain experts using the same quality rubric as real data.

Common Mistakes to Avoid

Mistake: Prioritising data volume over data quality for fine-tuning.

> **Correct approach:** For fine-tuning, quality dominates quantity. 1,000 high-quality, diverse, representative examples typically outperform 100,000 noisy, redundant, or misaligned examples. Audit your data quality before scaling data collection efforts.

Mistake: Assuming data licensing for one purpose covers AI training use.

> **Correct approach:** Many data licences explicitly prohibit AI training use, and this is increasingly enforced. Review AI training rights separately from general use rights. When in doubt, obtain explicit written confirmation from the data provider.

Mistake: Not holding out an evaluation set before training begins.

> **Correct approach:** If the evaluation set is curated after training (or from the same pool as training data), it may not provide an honest performance estimate. Create and freeze the evaluation set as the first step of any data project, before any model development begins.

Practitioner Tool: Fine-Tuning Data Curation Quality Checklist

Type: Checklist

Apply this checklist to a fine-tuning dataset before any model training begins. Each unchecked item represents a potential model quality or compliance risk.

- Data sources are documented with provenance (origin, date collected, licence confirmed)
- AI training rights explicitly confirmed for all data sources (not assumed from general licence)
- PII screening applied—personal names, emails, phone numbers, IDs removed or pseudonymised
- Exact deduplication applied—identical examples removed
- Near-deduplication applied—structurally similar examples with minor variation removed
- Length distribution reviewed—very short (< 50 tokens) and very long (> 2048 tokens) examples assessed
- Domain relevance verified—each example is representative of production query distribution
- Label quality audited—human review of at least 5% of labelled examples for annotation errors
- Bias assessment completed—performance across key demographic or topic subgroups evaluated
- Toxic content screening applied—examples containing harmful, illegal, or policy-violating content removed
- Copyright screening completed—examples reproducing copyrighted text at length flagged and reviewed
- Held-out evaluation set created and frozen before any training run begins
- Data version controlled—dataset is reproducible and linked to training run metadata

Note: This checklist should be signed off by both the technical lead and the responsible business owner before training commences. Incomplete items should be documented as accepted risks with mitigation plans.

Knowledge Check

Question 1: A team preparing a fine-tuning dataset for a customer service AI has 100,000 examples but has not reviewed data quality. A quick sample audit reveals 40% duplicates, 15% examples misaligned with the target task, and 5% containing customer PII. What is the correct next step?

- A. Proceed with training—100,000 examples is sufficient to absorb noise
- **B. Apply deduplication, relevance filtering, and PII removal before any training begins**
- C. Increase the training learning rate to make the model more robust to noisy examples
- D. Use the full dataset for training and rely on post-training evaluation to detect quality issues

Correct Answer: B. Apply deduplication, relevance filtering, and PII removal before any training begins

Explanation: Training on noisy, duplicate, or PII-containing data produces models with memorisation risk, poor generalisation, and privacy compliance failures. Data cleaning must precede training—post-training corrections are far more expensive than pre-training data curation.

Question 2: A fintech company wants to use a data provider's API responses as fine-tuning examples for a financial AI. The data provider's general licence allows commercial use. What additional step is required?

- A. No additional steps—commercial use licence covers AI training
- **B. Confirm the data provider explicitly permits AI training use, as many general commercial licences do not**
- C. Train the model privately without disclosing the data source
- D. Apply for a research exemption to use the data for AI training

Correct Answer: B. Confirm the data provider explicitly permits AI training use, as many general commercial licences do not

Explanation: Many data providers have updated their terms to explicitly prohibit AI training, even under commercial licences. Assuming general commercial rights cover AI training without explicit confirmation is a legal risk.

Question 3: A model fine-tuned on a company's customer interaction logs performs well in testing but shows significantly worse response quality for queries from non-native English speakers compared to native English speakers. What is the most likely root cause?

- A. The model's context window is shorter for non-standard English
- **B. Training data bias—non-native English speaker interactions are likely underrepresented or differently handled in the historical logs**
- C. Non-native English creates tokenisation errors that degrade model performance
- D. Fine-tuning learning rate was too high for language diversity

Correct Answer: B. Training data bias—non-native English speaker interactions are likely underrepresented or differently handled in the historical logs

Explanation: Performance disparity across demographic or language subgroups is a classic training data bias indicator. If non-native speaker interactions were less frequent in historical logs, or were handled differently (escalated to humans, closed faster), the model learns an imbalanced representation.

Reflection Questions

Take time to consider the following questions before the next session:

1. Web-scraped training data is economically accessible but introduces misinformation, bias, and copyright concerns. Fully licensed, curated data is expensive but cleaner. How should organisations balance these trade-offs in their AI strategy?
2. The 'garbage in, garbage out' principle implies that AI system quality is fundamentally a data governance problem. How should data governance practices evolve in organisations that are building or deploying AI?
3. Historical data reflects historical practices, which may include discriminatory decisions. Training AI on historical data perpetuates those patterns into the future. What frameworks exist for addressing historical bias in training data?
4. Synthetic data generated by AI is increasingly used to train other AI models ('model-generated data'). What are the risks of this feedback loop, and how should the field address them?
5. Data provenance and AI training consent are becoming regulatory requirements globally. What infrastructure investments should organisations make now to be compliant with upcoming regulations?

Key Takeaways

- Data quality is the primary determinant of model quality—architecture and compute can only work with what the training data contains.
- Pre-training requires trillions of tokens; fine-tuning requires thousands to hundreds of thousands of high-quality examples; RLHF requires preference pairs.
- High-quality training data must be accurate, diverse, relevant, consistent, and clean—removing duplicates, PII, toxic content, and copyright-infringing material.

- Bias in training data produces biased models: under-represented groups receive worse performance; historical discriminatory patterns are perpetuated.
- For fine-tuning, data annotation cost typically exceeds training compute cost by 10–100x—invest in data quality upfront rather than hoping to correct issues post-training.
- Data provenance documentation, AI training licence confirmation, and bias assessment are non-negotiable steps before any production training run.

Next Actions

Complete at least two of the following before the next lecture:

6. Audit one dataset you work with or have access to using the first five items of the Fine-Tuning Data Curation Quality Checklist—document what you find.
7. Review the EU AI Act's training data documentation requirements (Article 10) to understand what provenance records will be mandated for high-risk AI systems.
8. Identify one training dataset used by a model you deploy and check whether its Terms of Service explicitly permit AI training—document the finding.
9. Read a bias audit report from a deployed AI system (several are publicly available from organisations like the Algorithmic Justice League or NIST) and assess the methodology used.